

**346,232  
vulnerabilities.**

**Which ones deserve  
attention first?**

Machine-learning based vulnerability profiling & threat-intelligence workflow

CVSS

EPSS

CISA KEV

ATTACK

AGE

A capstone project exploring vulnerability profiles beyond severity-only analysis.

## THE PROBLEM

# Same severity. Very different threat profile.

CVSS is important - but severity alone does not capture exploitation likelihood, known exploitation, attack characteristics or age.

### CVE A

|      |     |
|------|-----|
| CVSS | 9.0 |
|------|-----|

|      |      |
|------|------|
| EPSS | 0.90 |
|------|------|

|          |     |
|----------|-----|
| CISA KEV | YES |
|----------|-----|

|               |         |
|---------------|---------|
| Attack Vector | NETWORK |
|---------------|---------|

### CVE B

|      |     |
|------|-----|
| CVSS | 9.0 |
|------|-----|

|      |      |
|------|------|
| EPSS | 0.01 |
|------|------|

|          |    |
|----------|----|
| CISA KEV | NO |
|----------|----|

|               |       |
|---------------|-------|
| Attack Vector | LOCAL |
|---------------|-------|

**VULNIQ asks: can multiple vulnerability characteristics reveal meaningful natural profiles?**

# I did not manufacture priority labels.

There was no reliable target such as enterprise remediation priority, so I used unsupervised learning to discover natural vulnerability profiles.

**NO**

**Artificial  
Critical / High /  
Medium / Low  
labels**

**YES**

**Unsupervised ML  
→ discover profiles  
from similarity**

## Algorithms evaluated

**K-MEANS**

Scalable centroid-based clustering

**AGGLOMERATIVE**

Hierarchical comparison on sample

**DBSCAN**

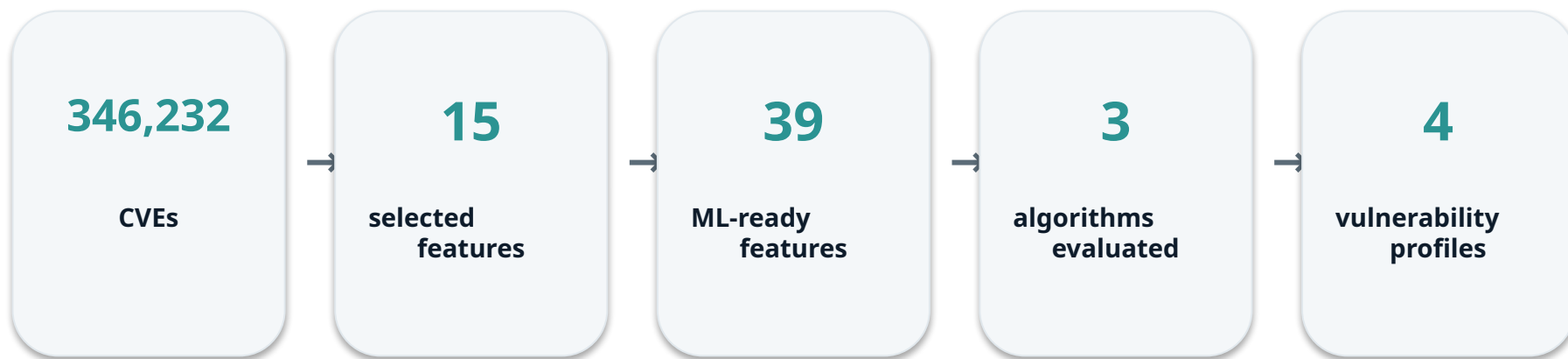
Density-based clustering + noise

**Selected profiling model: K-Means, K = 4**

## WHAT WE BUILT

# An end-to-end vulnerability profiling pipeline.

Every step below is part of the current VULNIQ capstone.



## FEATURES USED FOR CLUSTERING

**Severity & impact • EPSS • CISA KEV • Attack characteristics • Confidentiality / Integrity / Availability impact • Vulnerability age**

Preprocessing → feature engineering → encoding → scaling → clustering → evaluation → profiling

**Result: four interpretable vulnerability populations - not artificial risk labels.**

WHAT THE MODEL FOUND

1.76%

of CVEs



58.08%

of CISA KEVs

Cluster 0 contained 6,100 CVEs and 960 known exploited vulnerabilities.

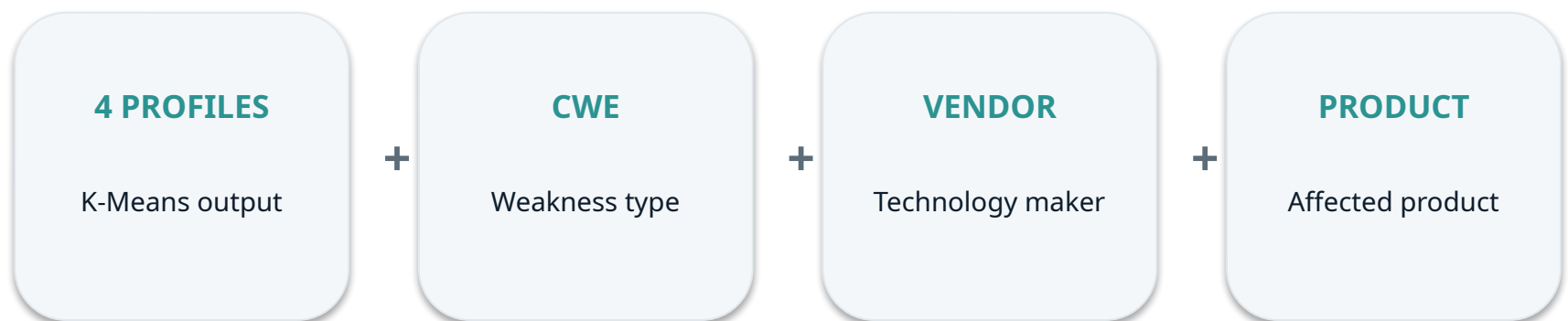
|           |                               |        |             |
|-----------|-------------------------------|--------|-------------|
| CLUSTER 0 | Exploitation Hotspot          | 1.76%  | 58.08% KEVs |
| CLUSTER 1 | High Severity & Impact        | 30.94% | 33.15% KEVs |
| CLUSTER 2 | Older & Highly Exploitable    | 20.17% | 0% KEVs     |
| CLUSTER 3 | Large Lower-Signal Population | 47.13% | 8.77% KEVs  |

KEV was included as a clustering feature; this concentration describes the resulting profile and is not independent predictive validation.

## WHAT ENRICHMENT ADDED

# From cluster IDs to understandable vulnerability profiles.

After clustering, the CVEs were enriched using a second dataset. CWE, vendor and product did not determine the clusters.



## WHAT THIS LET US UNDERSTAND

### Weakness patterns

Which CWE categories recur within each profile

### Technology concentration

Which vendors and products are common in each profile

### Remediation insight

Which vulnerability populations could be analyzed or managed together

**Clustering created the profiles. Threat-intelligence enrichment made those profiles easier to interpret.**

# From 346K vulnerabilities to interpretable profiles.

The project delivers a complete unsupervised ML workflow - from raw vulnerability data to profiled and enriched results.

|    |                                                   |                                                           |
|----|---------------------------------------------------|-----------------------------------------------------------|
| 01 | <b>346,232 CVEs analyzed</b>                      | Severity, exploitation, attack and impact characteristics |
| 02 | <b>3 clustering approaches evaluated</b>          | K-Means • Agglomerative • DBSCAN                          |
| 03 | <b>4 vulnerability profiles identified</b>        | Distinct populations rather than artificial risk labels   |
| 04 | <b>Exploitation concentration surfaced</b>        | 1.76% of CVEs contained 58.08% of CISA KEVs               |
| 05 | <b>Profiles enriched with threat intelligence</b> | CWE • Vendor • Product                                    |

VULNIQ transforms a large vulnerability population into interpretable profiles that can support more focused vulnerability analysis and remediation planning.